

GP487 Web Mapping Milestone Report

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Description of the Community, Issue, and User Group

Lethbridge is a medium-sized city in southern Alberta that depends on the Oldman River and its reservoir system for municipal water supply (“Spring”, 2025).

Water scarcity is a large issue in Lethbridge. In recent years, the region has experienced below-average mountain snowpack and precipitation, which has largely reduced spring and summer runoff into the Oldman River system (“Spring”, 2025). Consequently, the Oldman Reservoir (the main reservoir feeding Lethbridge) was only approximately 65% full in 2025 (“Spring”, 2025). The entire Oldman River Basin has also been provincially designated as Drought Stage 3: Severe Drought, and Lethbridge has implemented voluntary water restrictions to reduce water waste, especially in the hotter summer months (“Spring”, 2025). In the Prairie provinces as a whole, water scarcity is a large issue due to the high demands from agricultural irrigation (Dryden, 2021). Southern Alberta is particularly dry, with Lethbridge having 330 days of sunlight per year (“Location”, 2025). Throughout the province, 100% of available water sources have been allocated, causing water vulnerability in both municipalities and industries (Dryden, 2021).

Lethbridge also faces problems with water quality. Henderson Lake has been a point of concern due to the growth of freshwater cyanobacteria (also known as blue-green algae), which release toxins that are harmful to humans (Sibbet, 2025). In response, the city has used algaecides to reduce bloom levels (Sibbet, 2025). However, climate change factors are causing algal blooms to become more frequent and severe (“Climate”, 2025). Rising water temperatures accelerate bacterial growth, while blooms absorb sunlight and further warm the water, creating a feedback loop (“Climate”, 2025). Reduced freshwater runoff increases water salinity, which not only affects irrigation and drinking water quality but also allows salt-tolerant algae to spread inland (“Climate”, 2025). Reduced rainfall caused by prolonged droughts allows water bodies to retain nutrients for longer periods, which are then used by the algae (“Climate”, 2025).

Overall, climate change is a driving factor of both water scarcity and water quality issues in Lethbridge. This project will illustrate these relationships by presenting a time series comparison of key climate factors with indicators of water scarcity and water quality over the last decade. We will focus on the climate change factors of surface temperature, precipitation levels, and drought conditions, the water scarcity indicator of reservoir and river water levels, and the

water quality indicator of cyanobacterial bloom frequency. Temporally visualizing these trends will help demonstrate how climate change impacts multiple facets of Lethbridge's water system.

The user group for this project is residents of the City of Lethbridge. We plan to increase public awareness of the connection between climate change and local water issues. While climate change is a widely recognized issue in Canadian society, its local impacts can be less visible, making it difficult for residents to understand challenges within their own community. A web map is well-suited for this project topic and user group, since interactive web mapping makes information more accessible than static reports. Users can drive their own discovery of trends over different time periods, as well as focus on locations of particular interest, making conclusions more personally meaningful. By visualizing climate variables alongside water scarcity and quality indicators within the Lethbridge area, residents can directly see how environmental change intersects with the places that are important to them. Additionally, by publishing this project through an open online platform, it becomes more transparent and accessible. Users will be able to engage with this issue without needing to request datasets, attend council meetings, or rely on technical reports that might be difficult to understand. This open-access, localized approach will help residents better understand the reasoning behind voluntary water restrictions and support community engagement with city-wide water conservation plans.

Information Sources

Climate Change and Freshwater Harmful Algal Blooms. US Environmental Protection Agency.

(2025, February 26). <https://www.epa.gov/habs/climate-change-and-freshwater-harmful-algal-blooms>

Dryden, J. (2021, October 21). *Water shortages are a major risk of climate change. Alberta may already be seeing warning signs* | CBC News. CBC News.

<https://www.cbc.ca/news/canada/calgary/oldman-river-climate-change-jason-nixon-lethbridge-alberta-1.6214380>

Kim, H. G., Cha, Y., & Cho, K. H. (2024). Projected climate change impact on cyanobacterial bloom phenology in temperate rivers based on temperature dependency. *Water Research*, 249, 120928. <https://doi.org/10.1016/j.watres.2023.120928>

Location spotlight – Lethbridge, Alberta. Government of Canada. (2025, September 24).

<https://www.canada.ca/en/prairies-economic-development/corporate/contact/location-spotlight/lethbridge.html>

Sibbet, J. (2025, July 24). *City of Lethbridge taking proactive approach to blue-green algae.*

Global News. <https://globalnews.ca/news/11301287/city-of-lethbridge-blue-green-algae/>

Spring Water Update: Help Conserve in 2025. City of Lethbridge. (2025, May 23).

<https://www.lethbridge.ca/news/posts/spring-water-update-help-conserve-in-2025/>

Visser, P. M., Verspagen, J. M. H., Sandrini, G., Stal, L. J., Matthijs, H. C. P., Davis, T. W., Paerl, H. W., & Huisman, J. (2016). *How rising CO₂ and global warming may stimulate harmful cyanobacterial blooms.* Harmful algae, 54, 145–159. <https://doi.org/10.1016/j.hal.2015.12.006>

Data Sources

Category	Name and Link	Description
Background Info	City Boundary	City boundary of Lethbridge. Retrieved from the Lethbridge open data catalogue.
Background Info	Lakes and Rivers	Lakes and rivers in Lethbridge. Retrieved from the Lethbridge open data catalogue.
Background Info	Greenspace	Greenspace in Lethbridge. Retrieved from the Lethbridge open data catalogue.
Climate Change (temp)	Mean Temperature Difference From Normal	Mean surface temperature measurements in Canada. Retrieved from the Government of Canada.
Climate Change (temp)	Maximum Temperature	Maximum surface temperature measurements in Canada. Retrieved from the Government of Canada.
Climate Change	Minimum Temperature	Minimum surface temperature measurements in Canada. Retrieved from the Government of Canada.

(temp)		
Climate Change (precip)	Accumulated Precipitation	Precipitation measurements in Canada. Retrieved from the Government of Canada.
Climate Change (precip)	Precipitation Percentile	Percentile precipitation measurements in Canada. Retrieved from the Government of Canada.
Climate Change (drought)	Canadian Drought Monitor	Drought classifications in Canada. Retrieved from the Government of Canada.
Water Scarcity (level)	Oldman River Water Level	Daily water level measurements in the Oldman River. Retrieved from the Government of Canada.
Water Scarcity (level)	Oldman Reservoir Water Level	Daily water level measurements in the Oldman Reservoir. Retrieved from the Government of Canada.
Water Scarcity (level)	St. Mary Reservoir Water Level	Daily water level measurements in the St. Mary Reservoir. Retrieved from the Government of Canada.
Water Scarcity (level)	Waterton Reservoir Water Level	Daily water level measurements in the Waterton Reservoir. Retrieved from the Government of Canada.
Water Quality (algae)	Cyanobacterial blooms in Alberta recreational waters	Cyanobacterial bloom measurements in Alberta. Retrieved from the Alberta open data catalogue.
Water Quality (risk)	Surface water quality risk	Risk classification of water bodies in Alberta. Retrieved from the Alberta open data catalogue.
Basemap	Lethbridge Aerial Imagery 2023	Most up-to-date Aerial Imagery for Lethbridge. Retrieved from Lethbridge ERSI REST Services

Experience Builder Apps

Mehak – <https://experience.arcgis.com/experience/9162c0f090e741b3a6716d9476036b54/>

Sarah – <https://experience.arcgis.com/experience/aaec1e4de3e94c9fa1d79f3d35c9afe7>

Charles-

<https://experience.arcgis.com/experience/5591986148654bfa8540cffb3911a5ef?draft=true>

Eugene – <https://experience.arcgis.com/experience/a53054a3c4e143ee8a9fcee958bf6138/>

Leaflet JavaScript Apps

Mehak – <https://env-gis-srv1.uwaterloo.ca/w2026/gp487/group5/Mehak/Leaflet/index.html>

Sarah – <https://env-gis-srv1.uwaterloo.ca/w2026/gp487/group5/sarah/index.html>

Charles – <https://env-gis-srv1.uwaterloo.ca/w2026/gp487/group5/Charles/Final.html>

Eugene –

https://env-gis-srv1.uwaterloo.ca/w2026/gp487/group5/Eugene/data%20html/Leaflet_Eugene.html